Penalty regularization for the MPS pencil: derivations of the Tikhonov variants

`chladni` working notes

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These notes derive the regularizations measured in `notes/smooth_filter.md` and implemented as `reg="tikhonov"` and `reg="tikhonov_qr"` in `chladni.mps`, plus the second-order variant under test. Statements are marked as *proved* or *measured*; the measured ones cite the note's sections.

1 Setting

At a parameter λ the method of particular solutions assembles a boundary block $A_B \in \mathbb{R}^{m_B \times n}$ and an interior block $A_I \in \mathbb{R}^{m_I \times n}$, one column per basis function, and equilibrates so that every column of the stack

$$A = \begin{bmatrix} A_B \\ A_I \end{bmatrix} \in \mathbb{R}^{m \times n}, \qquad m = m_B + m_I,$$

has unit norm. The *tension* is

$$\sigma_1(\lambda) = \min_{c \neq 0} \frac{\|A_B c\|}{\|A_I c\|}, \tag{1}$$

the smallest generalized singular value of the pair (A_B, A_I). Write the SVD of the stack as $A = U\Sigma V^{\mathsf{T}}$ with $\Sigma = \operatorname{diag}(s_1 \geq \cdots \geq s_n \geq 0)$, and split U by rows as $U = \begin{bmatrix} U_B \\ U_I \end{bmatrix}$. After equilibration $\|A\|_F = \sqrt{n}$ (a column that underflowed to exact zeros keeps norm 0).

The basis is over-complete, so Σ is a smooth ramp that meets a flat floor of rounding noise at $s_j/s_1 \approx 10^{-16}$. Directions near that floor carry no signal; the whole question is how to keep them out of (1).

2 The invariance that rules out a weighting

Lemma 1 (proved)**.** *For any invertible $M \in \mathbb{R}^{n \times n}$ the generalized singular values of $(A_B M, A_I M)$ equal those of (A_B, A_I).*

Proof. The substitution $c = Mc'$ is a bijection of $\mathbb{R}^n \setminus \{0\}$, and

$$\frac{\|A_B M c'\|}{\|A_I M c'\|} = \frac{\|A_B c\|}{\|A_I c\|}.$$

The min–max characterization of every generalized singular value is therefore unchanged. □

A column weighting $w_j = f(s_j/s_1)$ is a right factor $M = \mathrm{diag}(w)$, so by Lemma 1 it changes nothing until some $w_j = 0$ exactly (measured: 4×10^{-15} change under a random invertible M). Truncation works *because* its projector is singular. A continuous regularization must therefore change the *objective*, not the coordinates: it must be a penalty.

3 Truncation, in whitened coordinates

Let $y = \Sigma V^{\mathsf{T}} c$, so that $Ac = Uy$ and (1) becomes $\min_y \|U_B y\| / \|U_I y\|$. Rank truncation at r restricts y to its first r coordinates: the pencil $(U_{B,1}, U_{I,1})$ with $U_1 = U[:, 1{:}r]$, and back map $c = V_1 \Sigma_1^{-1} y$. The threshold rule re-derives $r(\lambda) = \#\{j : s_j \geq \mathrm{rtol} \cdot s_1\}$ at every λ, and r is a step function: wherever a noisy s_j sits near the cutoff, rounding decides it, and the rank flickers. Measured: 63–88 changes per 201 samples at every half-width from 2×10^{-2} down to 2×10^{-10} (§3 of the note).

4 Order-1 Tikhonov

Definition 1. For $\tau > 0$,

$$\sigma_\tau(\lambda) = \min_{c \neq 0} \frac{\sqrt{\|A_B c\|^2 + \tau^2 \|c\|^2}}{\|A_I c\|}, \tag{2}$$

which is the smallest generalized singular value of $\left(\begin{bmatrix} A_B \\ \tau I \end{bmatrix}, A_I \right)$.

Whitened form. With $y = \Sigma V^{\mathsf{T}} c$ (assume $s_j > 0$; otherwise take limits), $\tau^2 \|c\|^2 = \tau^2 \left\|\Sigma^{-1} y\right\|^2 = \sum_j g_j^2 y_j^2$ with

$$g_j = \frac{\tau}{s_j}, \tag{3}$$

so (2) is the pencil $\left(\begin{bmatrix} U_B \\ \mathrm{diag}(g) \end{bmatrix}, U_I \right)$. A weight $g_j = 0$ leaves a direction alone; $g_j \to \infty$ forbids it, which is truncation. The corresponding filter factor is

$$f_j = \frac{1}{1 + g_j^2} = \frac{s_j^2}{s_j^2 + \tau^2},$$

a one-decade-wide smooth step centred at $s_j = \tau$. The augmented numerator has full column rank for every $\tau > 0$, and (2) is continuous in (A_B, A_I): no rank is chosen, so none can change.

Scaling. Scaling the whole pencil by α divides τ's effect by α, so a scale-invariant choice is relative:

$$\tau = \mathrm{rtol} \cdot \|A\|_F = \mathrm{rtol} \cdot \sqrt{n} \quad \text{(equilibrated)},$$

a constant for the pencil's life. The penalty acts in equilibrated coordinates, which is what makes $\|c\|$ meaningful: raw Fourier–Bessel columns span about 1149 nats of scale.

5 Two realizations: identical on paper, not in LAPACK

Literal (`"tikhonov"`). Hand the GSVD the pair $\left(\begin{bmatrix} A_B \\ \tau I \end{bmatrix}, A_I\right)$ directly. Back map: the identity.

QR (`"tikhonov_qr"`). Take an unpivoted QR

$$M_\tau = \begin{bmatrix} A_B \\ \tau I \\ A_I \end{bmatrix} = QR, \qquad Q = \begin{bmatrix} Q_1 \\ Q_2 \end{bmatrix},$$

with Q_1 the rows belonging to $\begin{bmatrix} A_B \\ \tau I \end{bmatrix}$, and hand the GSVD (Q_1, Q_2). Back map: $c = R^{-1}y$.

Proposition 1 (proved)**.** *The two realizations have identical generalized singular values in exact arithmetic.*

Proof. M_τ has full column rank because of the τI block, so R is invertible and $(Q_1, Q_2) = \left(\begin{bmatrix} A_B \\ \tau I \end{bmatrix} R^{-1},\ A_I R^{-1}\right)$. Apply Lemma 1 with $M = R^{-1}$. □

Why they differ in floating point (measured). τI makes the *stack* full rank but does nothing for the *interior* block. LAPACK's `?ggsvp3` decides the numerical rank of that block with its own tolerance, about $\max(m, n)\,\|A\|\,\varepsilon_{\text{mach}}$. So the literal form moves the rank decision inside LAPACK rather than removing it: its count of finite tensions ran 41–50 over a scan where the threshold took seven values. The QR form gives LAPACK an orthonormal stack, $Q_1^{\mathsf{T}}Q_1 + Q_2^{\mathsf{T}}Q_2 = I$, and the problem is then a CS decomposition with nothing to decide: the count held at n (or m_I) everywhere measured. The literal form's LAPACK cut is also why its floor stalls near 9×10^{-14}; the QR form's does not (§10 of the note).

6 Order 1: the floor, and why it cannot be traded against the noise

Proposition 2 (proved)**.** *Let c^* satisfy $\|A_B c^*\| = \epsilon\,\|A_I c^*\|$ (an eigenfunction resolved to ϵ by the basis), and let $\rho = \|c^*\| \,/\, \|A_I c^*\|$. Then*

$$\sigma_\tau \le \sqrt{\epsilon^2 + \tau^2\rho^2}.$$

Proof. Evaluate the quotient in (2) at c^*. □

Measured, the floor is 1.38–1.45 τ from $\tau = 1.7 \times 10^{-11}$ down to 1.7×10^{-14}, so $\rho \approx 1.45$; it rises to $1.9\,\tau$ at $\tau = 1.1 \times 10^{-15}$, where the basis's own ϵ enters, exactly as the bound says. That bottom value, 2.1×10^{-15}, is within 1.3× of the threshold's 1.6×10^{-15}. The eigenfunction lives in well-conditioned directions, yet it still pays $\tau\,\|c^*\|$: the order-1 penalty charges every direction, the signal included.

Noise (heuristic, and measured). Assembly and the QR are backward stable, so each is exact for a perturbation δA with $\|\delta A\| \approx \varepsilon_{\text{mach}} \|A\|$. In a noise direction the only thing holding the minimizer off it is the penalty, of size τ, against rounding of size $\varepsilon_{\text{mach}} \|A\|$. As τ falls toward $\varepsilon_{\text{mach}} \|A\| \approx 10^{-15}$, a weak penalty lets the minimizer lean on directions whose content is rounding. Measured: relative noise in σ_τ is 1×10^{-2} at the τ a 5×10^{-15} floor needs, against the threshold's 1×10^{-3}, and it is *multiplicative*, uniform along a well's flank. Floor and noise are both set by τ against $\varepsilon_{\text{mach}} \|A\|$, so no single τ makes both small.

7 Order 2: a penalty that spares the signal

The fix is to make the penalty depend on the direction: tiny on well-conditioned directions, large on noise. In whitened coordinates this is $g_j = (\tau_2/s_j)^2$ in place of (3). The point is that it can be built from triangular factors, without an SVD.

Lemma 2 (proved). *If $A = Q_0 R_0$ is an economic QR with R_0 invertible, then $\left\|R_0^{-\mathsf{T}} c\right\| = \left\|\Sigma^{-1} V^{\mathsf{T}} c\right\|$ for all c.*

Proof. $R_0^{\mathsf{T}} R_0 = A^{\mathsf{T}} A = V \Sigma^2 V^{\mathsf{T}}$, so $R_0 = W \Sigma V^{\mathsf{T}}$ for some orthogonal W (its singular values are Σ and its right singular vectors V). Hence $R_0^{-\mathsf{T}} = W \Sigma^{-1} V^{\mathsf{T}}$, and W preserves norms. □

The bounded form. R_0 is singular when a column underflows, and near-singular always. Use instead the factor R_1 of the τ_1-augmented stack, which `"tikhonov_qr"` already computes:

$$\begin{bmatrix} A_B \\ \tau_1 I \\ A_I \end{bmatrix} = Q_1' R_1, \qquad R_1^{\mathsf{T}} R_1 = A^{\mathsf{T}} A + \tau_1^2 I = V(\Sigma^2 + \tau_1^2 I) V^{\mathsf{T}}.$$

By the argument of Lemma 2, $\left\|R_1^{-\mathsf{T}} c\right\| = \left\|(\Sigma^2 + \tau_1^2 I)^{-1/2} V^{\mathsf{T}} c\right\|$, so $\left\|R_1^{-\mathsf{T}}\right\| \le 1/\tau_1$: bounded, and invertible whatever A does.

Definition 2. The order-2 penalized tension is the smallest generalized singular value of

$$\left(\begin{bmatrix} A_B \\ \tau_1 I \\ \tau_2^2 R_1^{-\mathsf{T}} \end{bmatrix}, \; A_I \right), \tag{4}$$

realized, like `"tikhonov_qr"`, by an unpivoted QR of the numerator stacked on A_I and a GSVD of the two row blocks of its Q.

Whitened form. With $y = \Sigma V^{\mathsf{T}} c$, $\tau_2^4 \left\|R_1^{-\mathsf{T}} c\right\|^2 = \sum_j \left(\tau_2^2/(s_j \sqrt{s_j^2 + \tau_1^2})\right)^2 y_j^2$. Together with the $\tau_1 I$ rows, direction j carries the weight

$$g_j^2 = \left(\frac{\tau_1}{s_j}\right)^2 + \left(\frac{\tau_2^2}{s_j \sqrt{s_j^2 + \tau_1^2}}\right)^2 \approx \begin{cases} (\tau_2/s_j)^4 & \tau_1 \ll s_j \quad \text{(order 2)}, \\ (\tau_2^2/(\tau_1 s_j))^2 & s_j \ll \tau_1 \quad \text{(order 1, bounded)}, \end{cases} \tag{5}$$

taking $\tau_1 \ll \tau_2$ so that the first term is negligible wherever it matters. In basis coordinates the block $\tau_2^2 R_1^{-\mathsf{T}}$ is dense lower-triangular, not diagonal. It is diagonal only in the SVD basis, and Lemma 2 is what lets us apply it without forming that basis.

Floor. Proposition 2 carries over with the penalty term replaced:

$$\sigma \le \sqrt{\epsilon^2 + \tau_1^2\rho^2 + \tau_2^4 \left\|(\Sigma^2+\tau_1^2)^{-1/2}V^{\mathsf{T}}c^*\right\|^2 / \|A_I c^*\|^2}.$$

If c^* lies in directions with $s_j \ge s_* \gg \tau_2$, the last term under the root is at most $(\tau_2^2\rho/s_*)^2$, and $\tau_2^2\rho/s_* \ll \tau_2\rho$. The signal is charged by $(\tau_2/s_j)^2$, not τ_2/s_j, so the floor can sit far *below* τ_2.

Noise (heuristic). In a noise direction ($s_j \approx \varepsilon_{\text{mach}}\|A\|$) the weight is $(\tau_2/s_j)^2$ rather than τ/s_j: the margin over rounding is the *square* of the order-1 margin at the same τ.

Measured (§10 of the note; family A, $n = 126$, $\tau_1 = 10^{-16}\sqrt{n}$):

penalty	worst floor	median floor	plateau noise
threshold, rtol $= 1.49 \times 10^{-15}$	1.6×10^{-15}	1.2×10^{-15}	1.0×10^{-3}
order 1 (QR), $\tau = 3.4 \times 10^{-15}$	5.2×10^{-15}	4.8×10^{-15}	1.0×10^{-2}
order 2, $\tau_2 = 3 \times 10^{-14}$	7.1×10^{-15}	4.3×10^{-15}	3.7×10^{-5}
order 2, $\tau_2 = 10^{-14}$	4.1×10^{-15}	2.2×10^{-15}	3.2×10^{-4}

At a matched $\approx 5 \times 10^{-15}$ floor, order 2 is about 270× quieter than order 1 and 27× quieter than the threshold. On the one prec $= 10^{-14}$ decline-grid cell checked so far ($145\pi^2$, band 1.3), the threshold declines on "the slope estimate never settled" and order 2 brackets all ten levels, pure, in 280 evaluations.

8 Relation to the hybrid

Section 6b of the note measured a "hybrid": an SVD of the stack, a hard cut below $\varepsilon_{\text{mach}} s_1$, and $g_j = (c\,s_1/s_j)^p$. At $p = 2$ that is (5) with an explicit SVD and a hard cut. Order 2 recovers the same filter shape with two QRs and one triangular inverse, and without either the SVD or a cut that a singular value could cross.

Summary

mode	penalty, basis coords	weight g_j, whitened	SVD?	rank decision?
threshold	— (truncate)	0 or ∞ at rtol s_1	yes	yes, flickers
`tikhonov`	τI	τ/s_j	no	yes, inside LAPACK
`tikhonov_qr`	τI (via QR)	τ/s_j	no	none
order 2	$\tau_1 I$, $\tau_2^2 R_1^{-\mathsf{T}}$	(5)	no	none
hybrid ($p = 2$)	SVD-built	$(c\,s_1/s_j)^2$, cut	yes	at $\varepsilon_{\text{mach}} s_1$